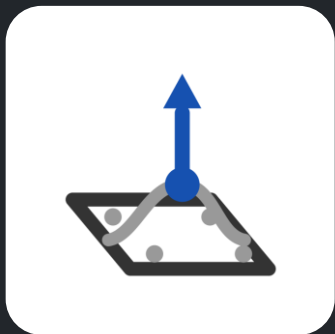




Pull surface

Push and pull a face with a grid of handles; the edges keep G0, G1 or G2 with their neighbours along their whole length.



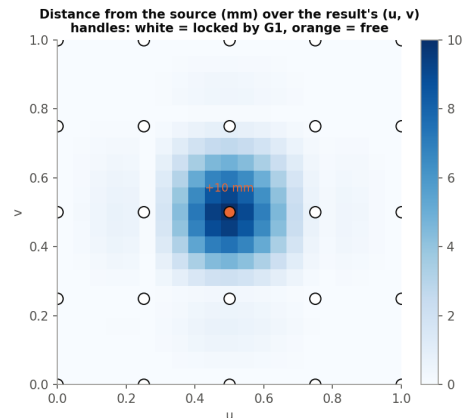
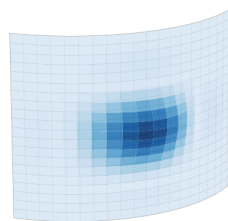
What it does

Onshape has no control-net push/pull. Pull surface puts arrow handles on a face; drag them and a new sheet is built from the face's own control net, with the edge rows locked so it still fits its neighbours.

Use it when

- Add a bump or dip to a face
- Tweak a surface without rebuilding its curves
- Keep continuity with surrounding faces

P2: source cylinder patch (grey net), pulled sheet coloured by distance from it

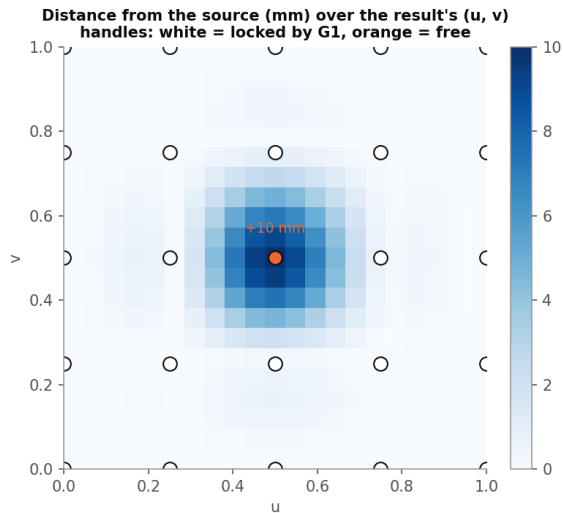
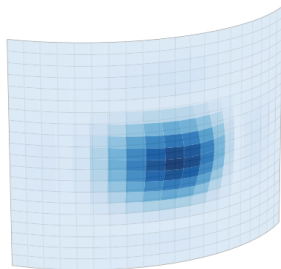


P2: +10 mm at the centre of a cylinder patch, G1 edges.



Locks and handles

P2: source cylinder patch (grey net), pulled sheet coloured by distance from it



Locked rows

1 / 2 / 3 control-point rows per edge for G0 / G1 / G2: continuity holds along the whole edge.

Handles

Free control points move along the normal; the offsets are solved so the surface passes through each handle.

Enough handles

G1 needs 5+ handles each way before any is free, G2 7+.



The dialog

IMAGE MISSING:
docs/decks/pull_surface/shots/dialog.png

#	Parameter	What it means
1	Face	The face to pull (kept; the result is a new sheet).
2	U curve count	Handles across u, edges included (2-20, 4).
3	V curve count	Handles across v.
4	Continuity	G0 / G1 / G2 along the edges.
5	Active offsets	U, V, Offset per handle; filled by dragging; 0 removes.
6	Debug	Iso-curves, keep curves / points, handle grid, offset vectors, print.
7	Approximation	Degree (minimum, up to 5); how exactly the face is read.



Outputs and messages

Published for Extract variables

Key	What it holds
output	The pulled sheet (pullSurf).

Messages

	Message / what to do
Info	no handle is free / holes not kept / solve fell back / trimmed edges held approximately One line.



Tips and limits



Counts first

Changing a count or the continuity resets the offsets; set them before dragging.



Trims

Holes are dropped; edges trimmed inside the face hold continuity only approximately.



Where it lives

Document	gordonSurface (d0950ed72a894fbe35c27d43)
Tab	pullSurface
Tests	Gordon tools tests: P1-P3
Examples	Test Part Studio 1: Pull surface 1
Explainer	docs/explainers/gordon_surface/gordon_surface_explained.md , section 2.5
Related	Gordon Surface , Interior curves , Scaled Curve , Modify curve end , Pull surface , Simplify surface